

WIRESHARK : THE BASICS

Wireshark is an open-source, cross-platform network packet ~~network~~ analyser tool capable of sniffing and investigating live traffic and inspecting packet captures (PCAP).

Wireshark is not an intrusion detection system (IDS). It only allows analysts to discover and investigate the packets in depth not modify them.

- Packet details are shown in three different panes:
 - a) Packet list pane: Summary of each packet (source ip, destination ip, protocol, and packet info)
 - b) Packet Details pane: Detailed protocol breakdown of the packet
 - c) Packet bytes pane: Hex and decoded ASCII representation of the selected packet.
- How to view default permanent colour settings?
View → coloring rules
- Traffic Sniffing: Also called as Packet Capturing
Blue shark button → Start network sniffing
Red button → Stop " "
Green button → Restart " "
- Merge PCAP files:
File → Merge

It will merge the existing pcap file with the chosen one and create a new pcap file.

- How to view file details like file hash, capture time, capture file comments, interface and statistics?
Statistics → Capture file properties
- Packet dissection: Also known as protocol dissection which investigates packet details by decoding available protocols and fields
- ~~Go to packet~~ Go-to packet: Go → Go to packet
- Find-Packets: Wireshark can find packets by packet content apart from packet number.

Edit → Find Packet

There are two crucial points in finding packet: First is knowing the input type which are (String, Hex, Regex, Display filter). Searches are case insensitive.

Second is choosing the search field like where you want to search like in packet list, packet details, packet bytes.

- Mark packets: Marking packets can be another helpful functionality. It can be done by right clicking on a packet and select mark
The marked packets information is removed every file session, so marked packets will be lost after closing the capture file
- Packets comments: You can add comments for particular packets that will help the further investigation by right clicking the packet and select packet comment. Unlike packet marking, the comment can stay within the capture file until operator removes them.

- **Export packets:** This functionality helps analysts to share only suspicious packets.

File → Export specified packets

- **Export objects (files) →** Wireshark can extract files transferred through wire. Exporting objects are available only for selected protocol's stream (DICOM, HTTP, IMF etc)

- **Time Display format:** Wireshark lists packets 'Seconds since beginning of capture'. You can use the UTC Time display format for a better view.

View → Time display format

- **Packet filtering:** Wireshark has two types of filtering approaches capture and display filters.

Capture filters are used for capturing only the packets valid for used filter, and display filters are used for 'viewing' the packets valid for the used filter.

- **Apply as filter:** Analyse → Apply as filter

While investigating a capture file you can click on the field you want to filter and use the 'right click menu' to filter the specific value.

- **Conversation filter:** Suppose you want to investigate a specific packet number and all linked packets ~~and~~ by focusing on IP address and port numbers.

Analyse → Conversation filter

- Colorize conversation: It highlights the linked packet without applying a display filter and reducing the number of viewed packets.

View → Colorize conversation

How to reset? View → ~~Conversation~~ ~~Filtered~~ Colorize conversation → Reset Colorization

- Prepare as filter: Unlike apply as filter this model doesn't apply the filters after the choice. It adds the requested query to the pane and wait for the execution command or another choosing filtering option by using and/or.
- Apply as column: Analyse → Apply as column
It helps you add a value and apply it as column and then it will be visible in packet list pane.
- Follow stream: Wireshark displays everything in packet portion size. However it is possible to reconstruct the streams and view the raw traffic as it is presented at the application level.

Analyse → Follow TCP/UDP/HTTP stream